BONES

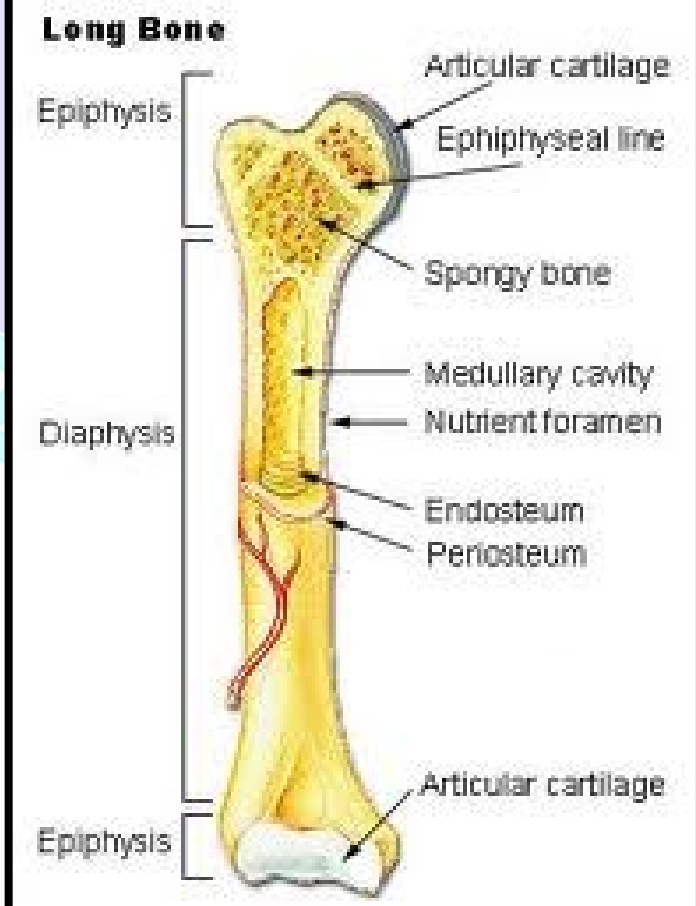
DIVISION AND FUNCTION OF SKELETAL SYSTEM

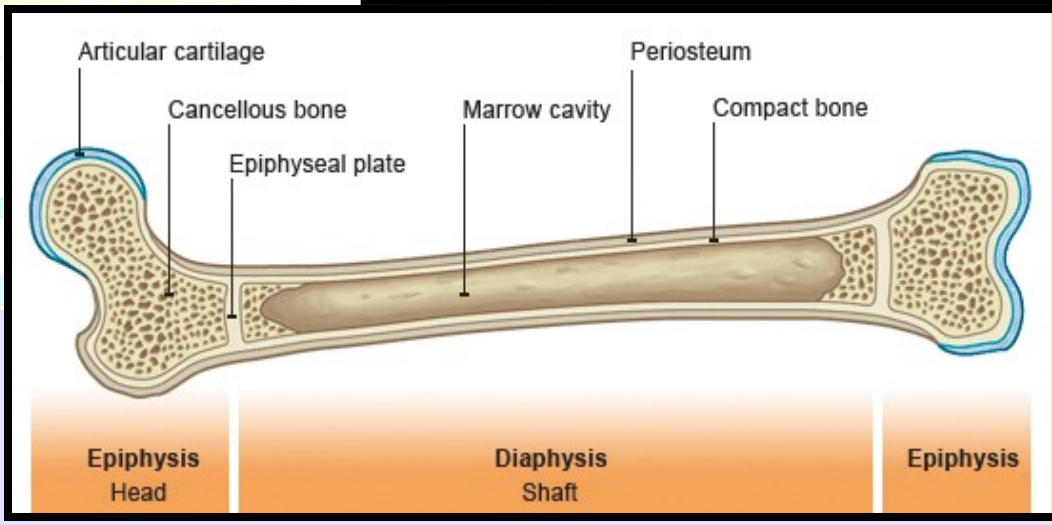
LEARNING OBJECTIVES

- **At the end of the lecture the student should be able to:**
  - Explain the division and function of skeletal system
  - Classify bones according to shape and regions.
  - Identify the parts of adult long bone.
  - Identify the parts of young long bone.

BONE

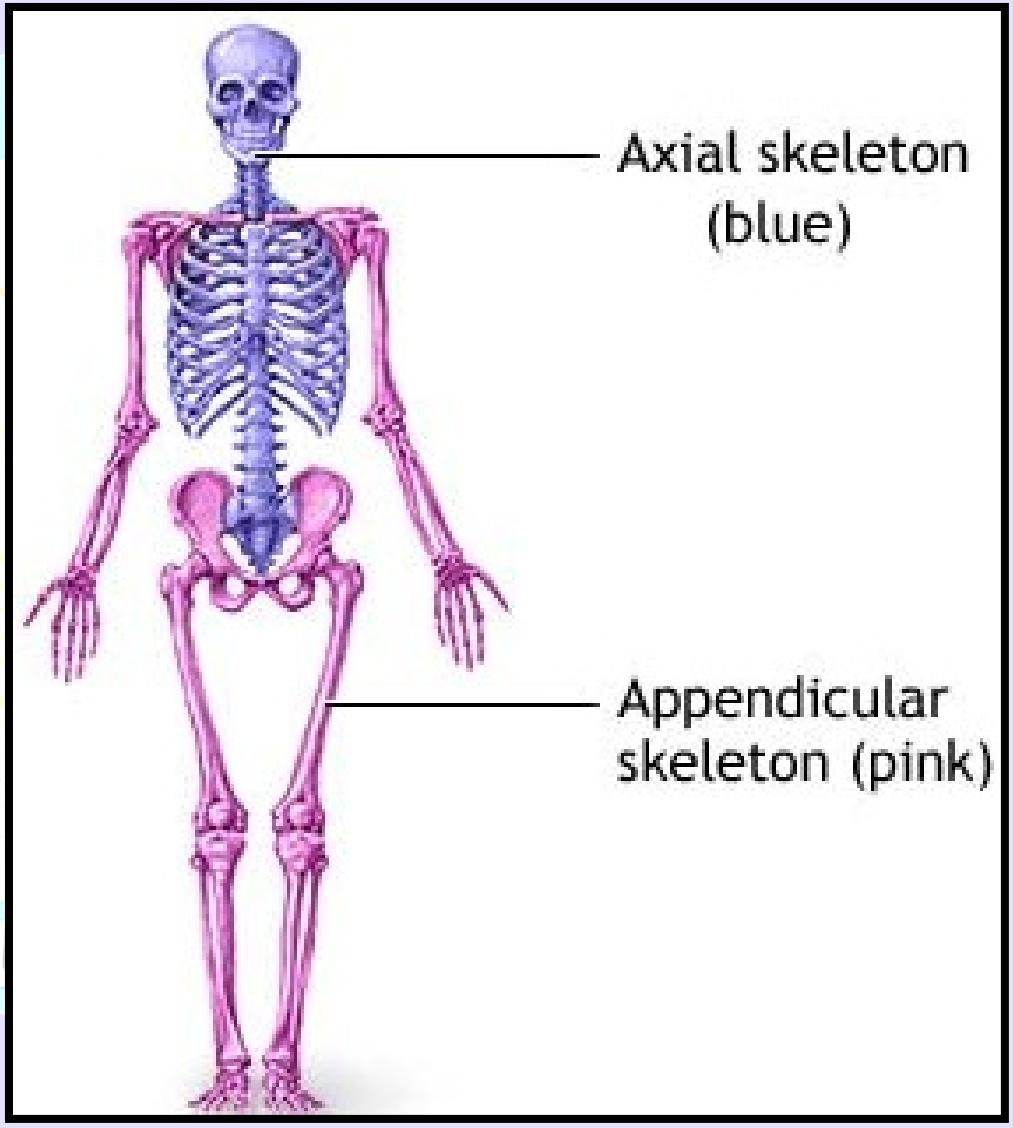
- Hardest structures of the animal body.
- Its color, in a fresh state, is pinkish-white externally, and deep red within.



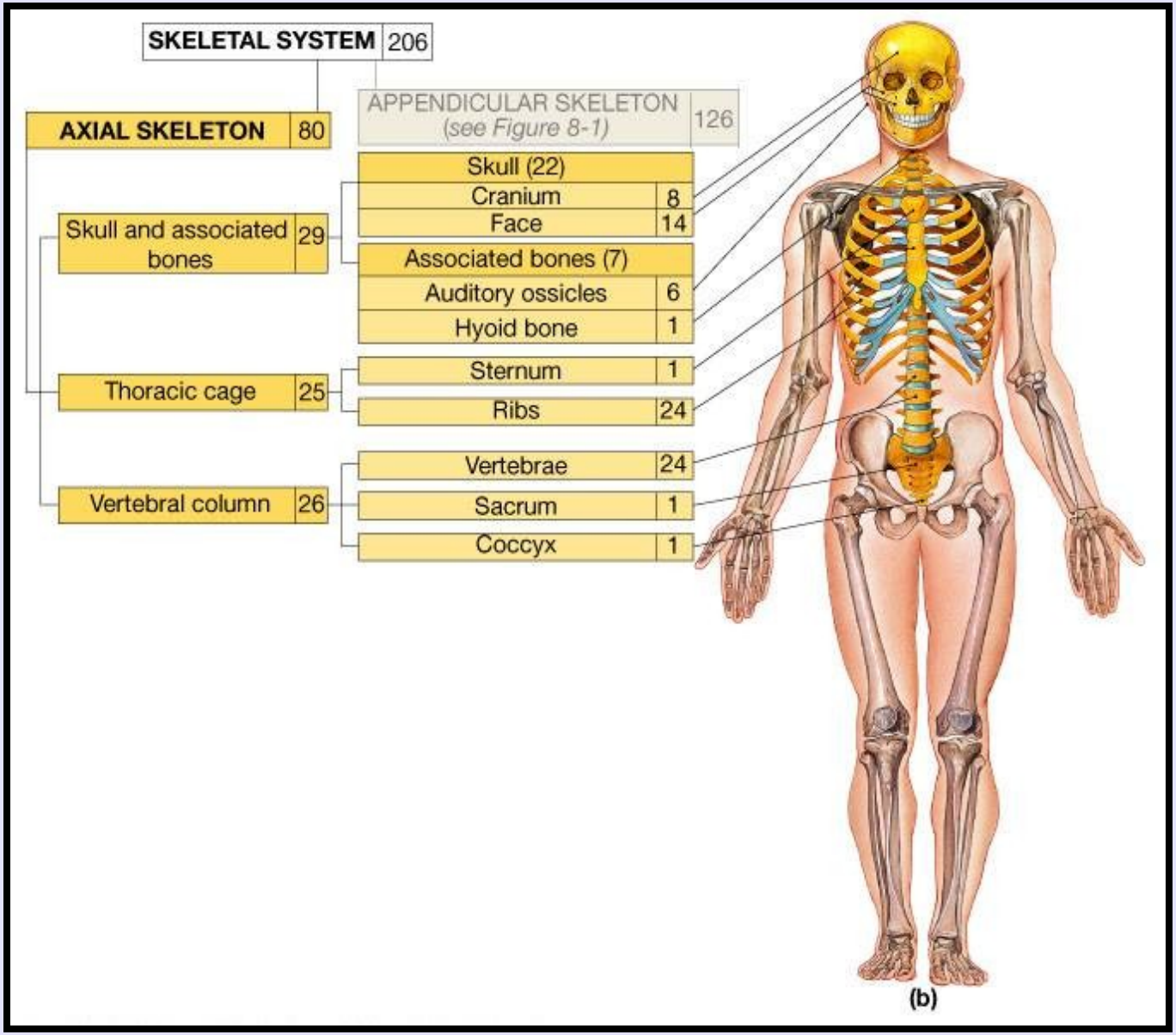


REGIONAL CLASSIFICATION

- **Axial** skeleton includes
  - skull,
  - vertebral column
  - thoracic cage
- **Appendicular** skeleton includes
  - bones of the limbs.

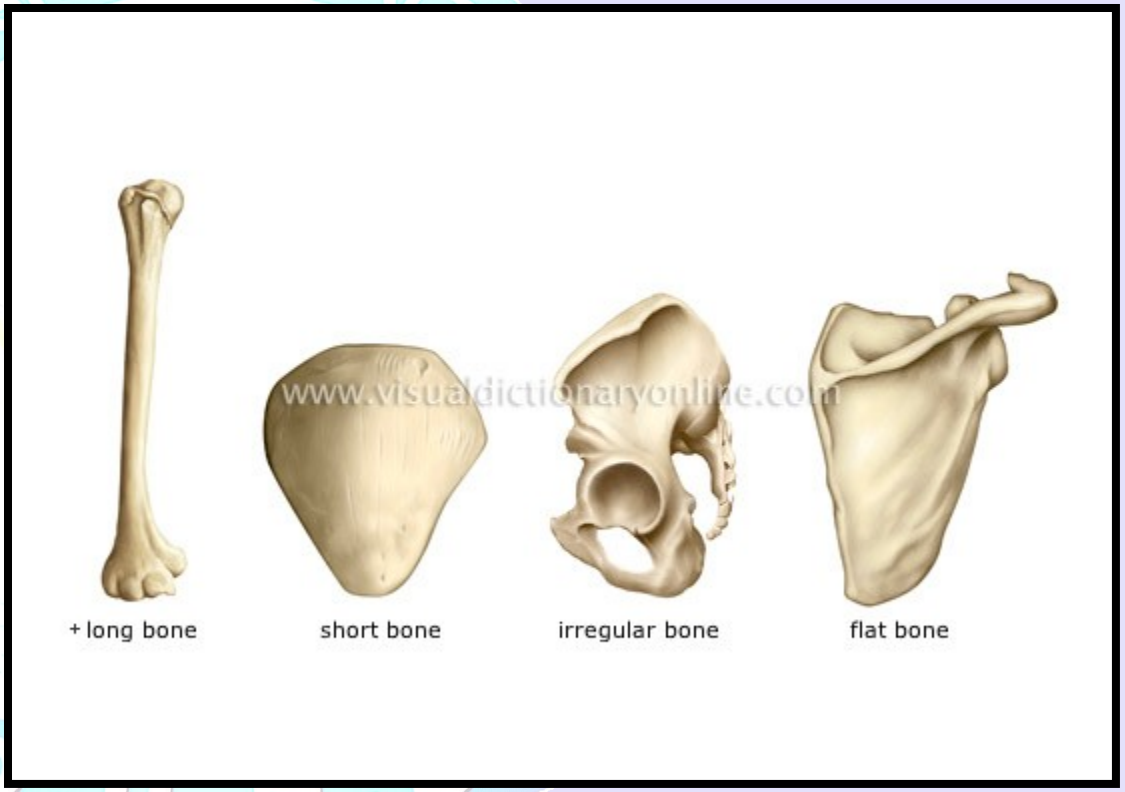


REGIONAL CLASSIFICATION



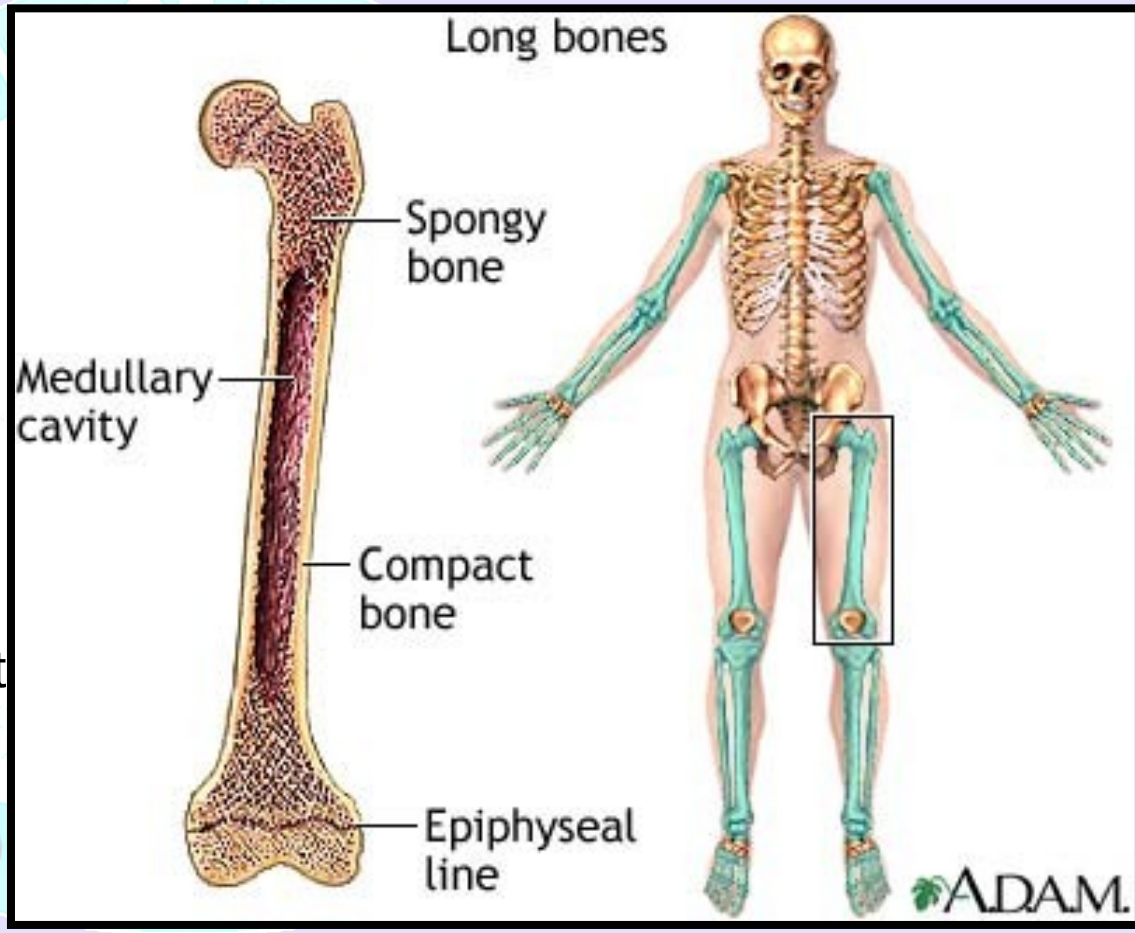
Classification According to shape

- There are SIX types of bones in the human body:
  - ***Long***
  - ***Short***
  - ***Flat***
  - ***Irregular***
  - ***Sesamoid***
  - ***Pneumatic***.



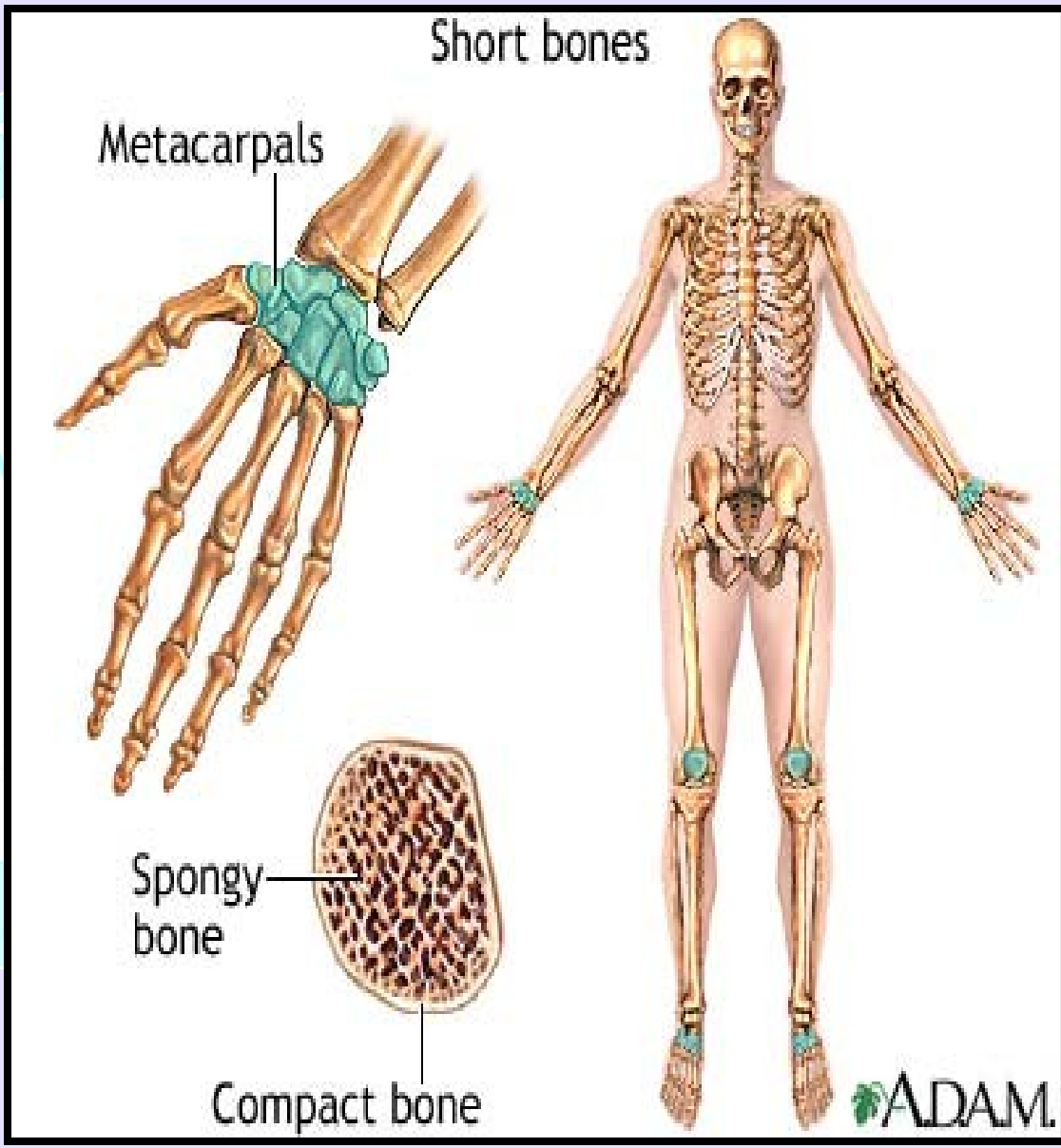
Long bones

- **Long bones** are longer than they are wide, consisting of a
- long shaft (the ***diaphysis***)
- Two articular (joint) surfaces called ***epiphyses.***
- Most bones of the limbs (including the three bones of the fingers) are long bones, except for the kneecap (patella), and the ***carpal, metacarpal***, ***tarsal*** and ***metatarsal*** bones of the wrist and ankle ***(miniature long bones).***
- The classification refers to shape rather than the size.



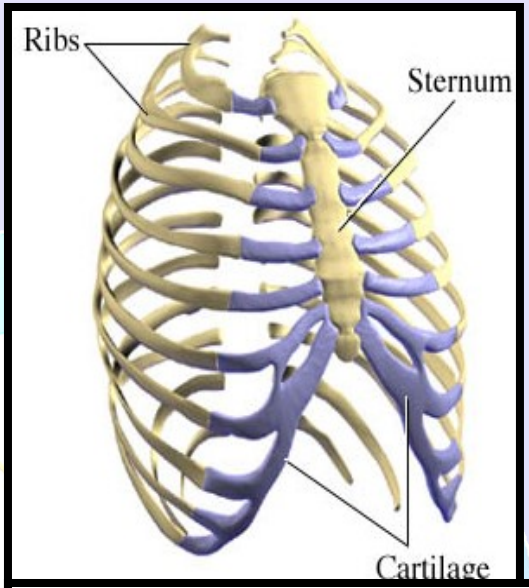
SHORT BONES

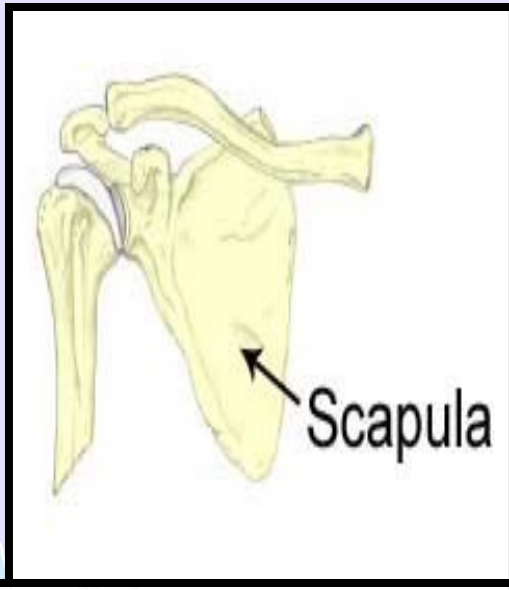
- **Short bones** are roughly cube-shaped, and have only a thin layer of compact bone surrounding a spongy interior.
- The bones of the wrist and ankle e.g. cuboid cuneiform, trapezoid, or scaphoid i.e. carpal and tarsal bones are short bones.

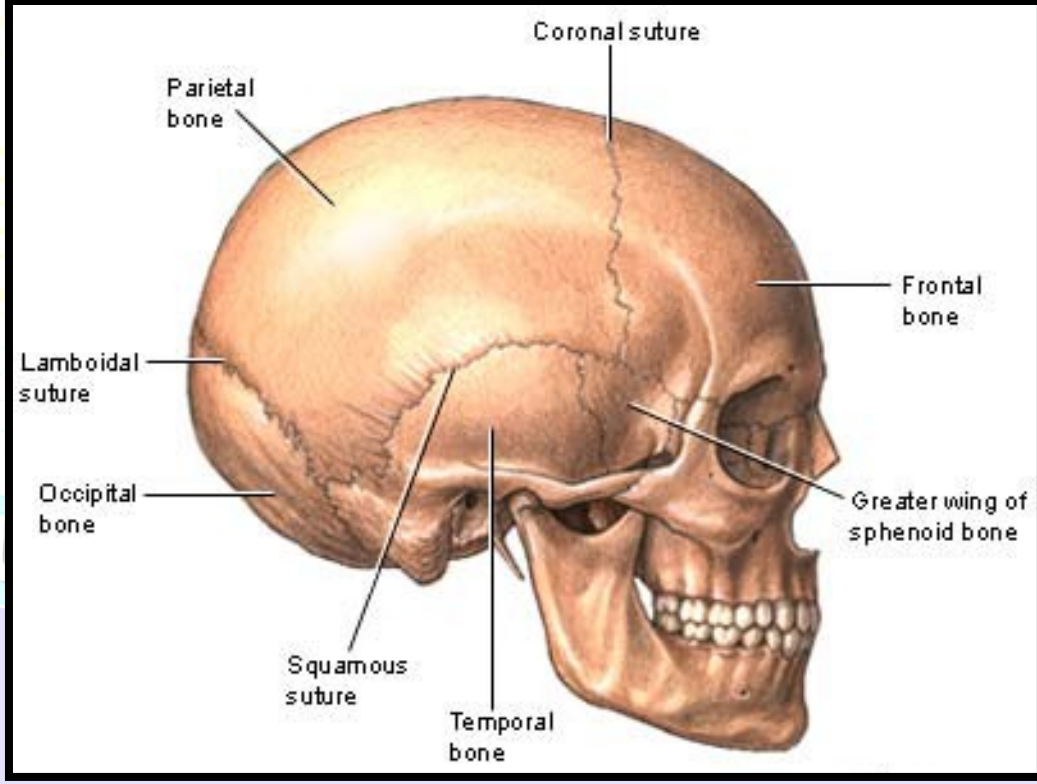


FLAT BONES

- **Flat bones** are thin and generally curved, with two parallel layers of compact bones sandwiching a layer of spongy bone.
- Most of the bones of the ***skull*** are flat bones, as is the ***sternum***.
- They resemble shallow plates and form boundaries of certain body cavities, Example: bones in the vault of the skull, ***ribs***, sternum and ***scapula***.

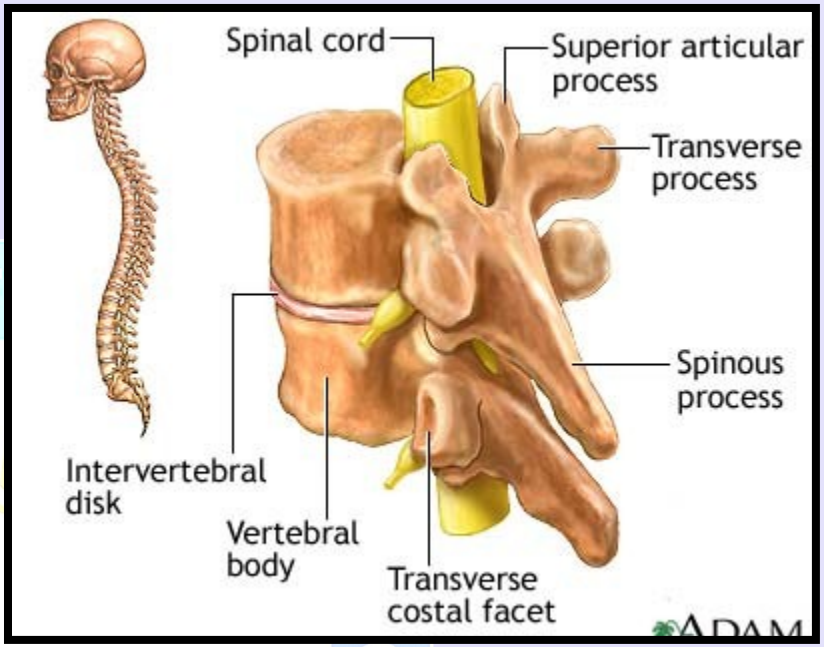






IRREGULAR BONES

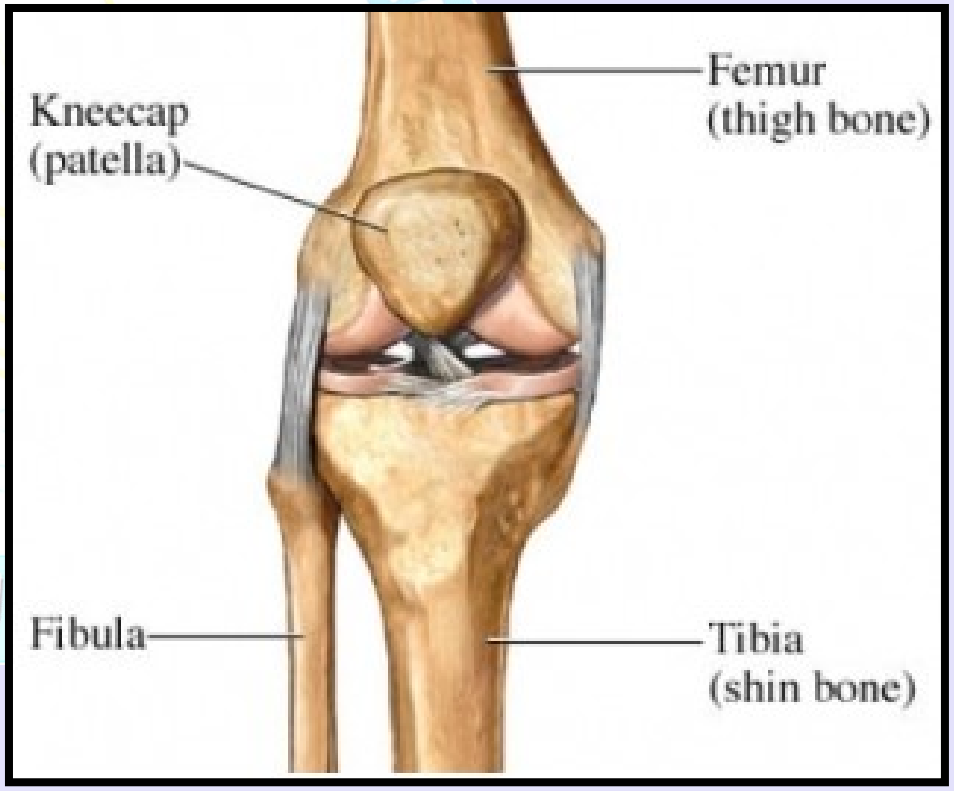
- **Irregular bones** do not fit into the above categories. They consist of thin layers of compact bone surrounding a spongy interior. As implied by the name, their shapes are irregular and complicated.
- The bones of the ***spine (vertebrae)***, bones in the ***base of the skull*** *and* ***hips*** are irregular bones.





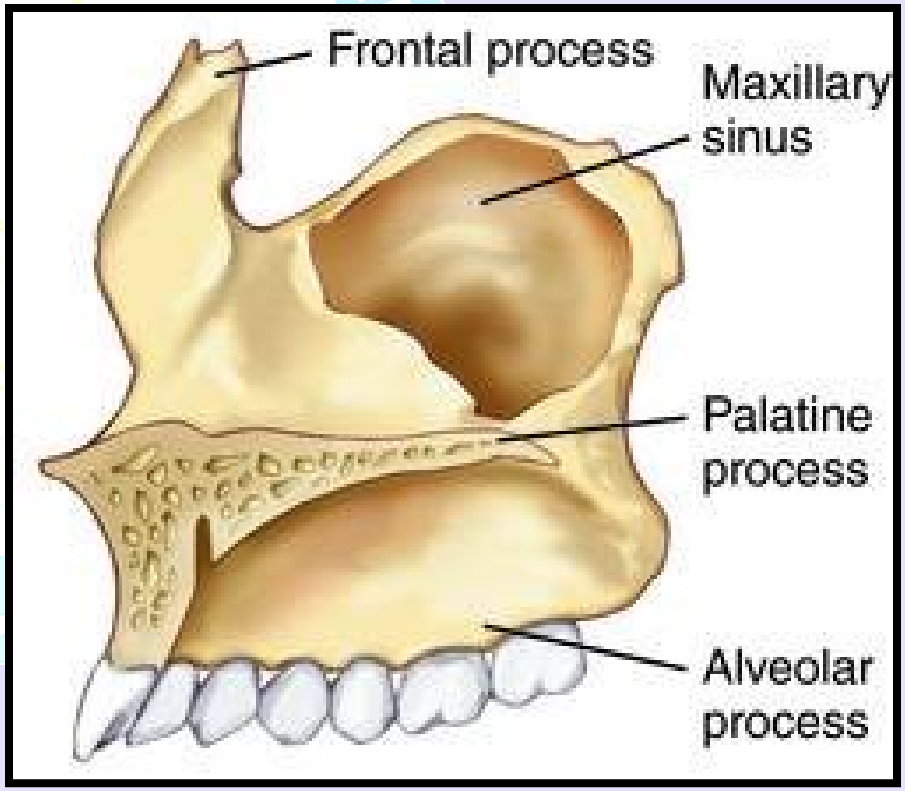
SESAMOID BONES

- **Sesamoid bones** are bones embedded in tendons or joint capsules.
- They have no periosteum, ossify after birth.
- Examples of sesamoid bones are the ***Patella*** and the ***Pisiform***
- **Functions**
  - Resist pressure
  - Minimize friction
  - Alter the direction of pull of the muscle and
  - Maintain the local circulation.



PNEUMATIC BONES

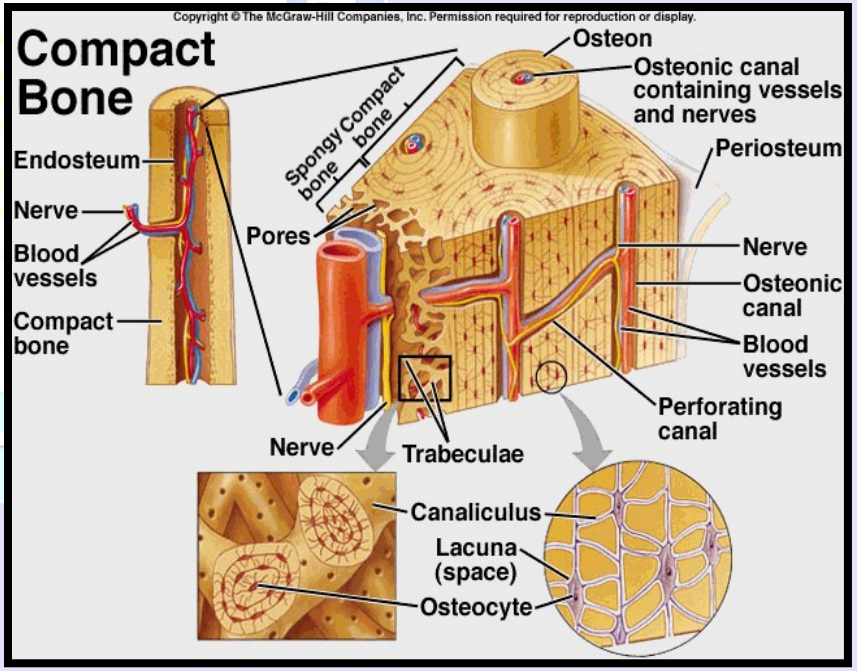
- Certain irregular bones contain large air spaces lined by epithelium.
- Example: maxilla, sphenoid, ethmoid etc.
- They make the skull light in weight, help in resonance of voice, and act as air conditioning chambers for the inspired air



(STRUCTURAL CLASSIFICATION)

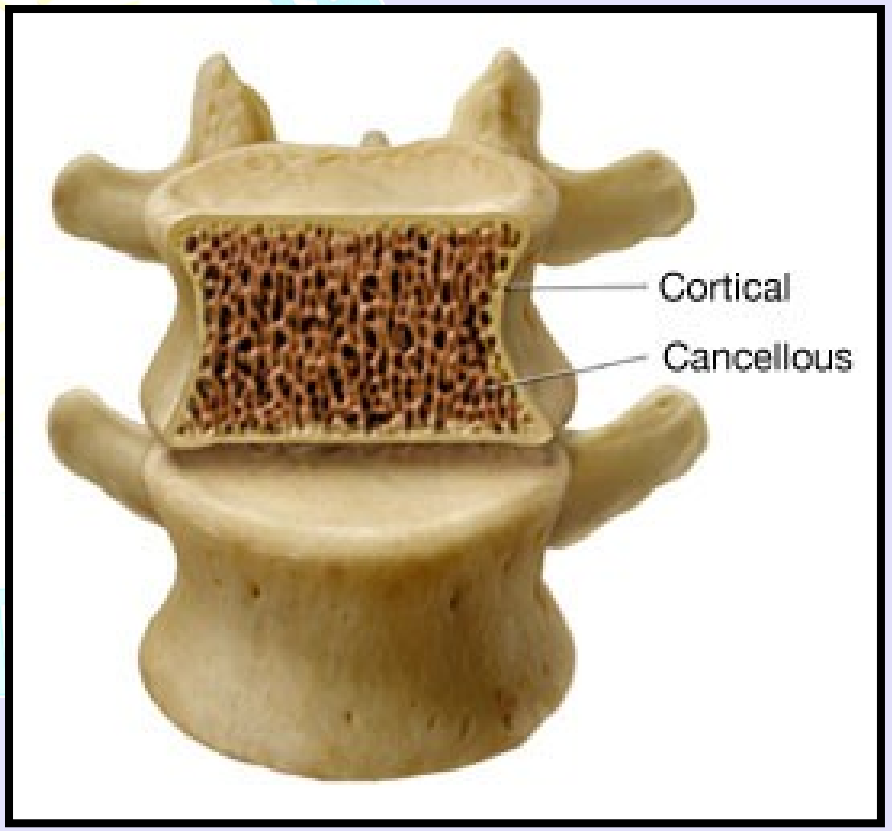
Macrostructure

- Bone is not a uniformly solid material, but rather has some spaces between its hard elements.
- **Compact bone (cortical bone)**
- The hard outer layer of bones is composed of ***compact bone*** tissue.
- Has less gaps and spaces.
- This tissue gives bones their smooth, white, and solid appearance, and
- Accounts for 80% of the total bone mass of an adult skeleton.



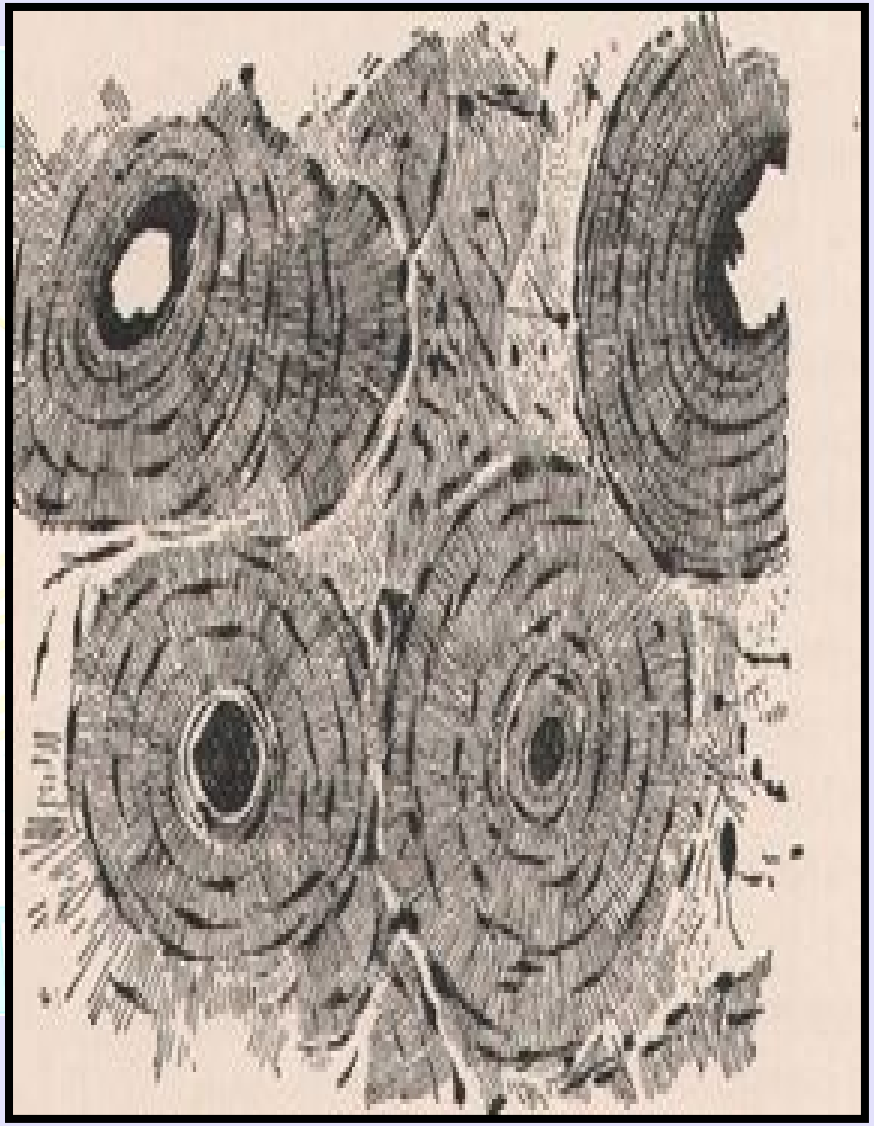
TRABECULAR BONE

- Filling the interior of the organ is the ***trabecular bone*** tissue (an open cell ***porous*** network also called ***cancellous or spongy bone***)
- Consists of a network of rod- and plate-like elements that make the overall organ lighter and allowing room for blood vessels and marrow.
- Accounts for the remaining 20% of total bone mass, but has nearly ten times the surface area of compact bone



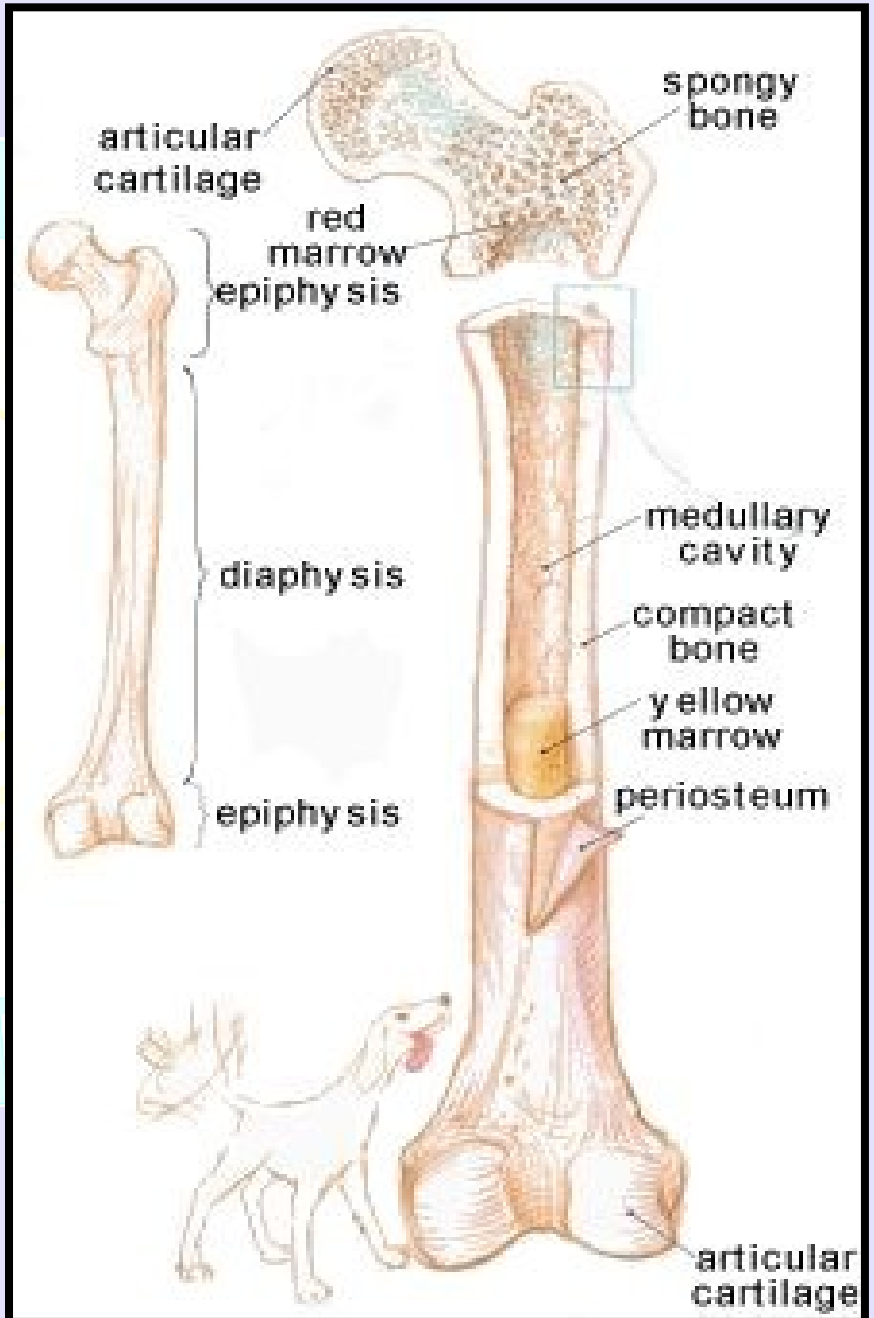
WOVEN AND LAMELLAR

- Bone can be either
- **Woven**
- This bone is weak, with a small number of randomly oriented collagen fibers, but forms quickly and without a pre-existing structure during periods of repair or growth
- **Lamellar (layered).**
- Lamellar bone is stronger, formed of numerous stacked layers and filled with many collagen fibers parallel to other fibers in the same layer. The fibers run in opposite directions in alternating layers, assisting in the bone's ability to resist torsion forces.
- After a break, woven bone quickly forms and is gradually replaced by slow-growing lamellar bone on pre-existing calcified ***hyaline cartilage*** through a process known as "bony substitution."



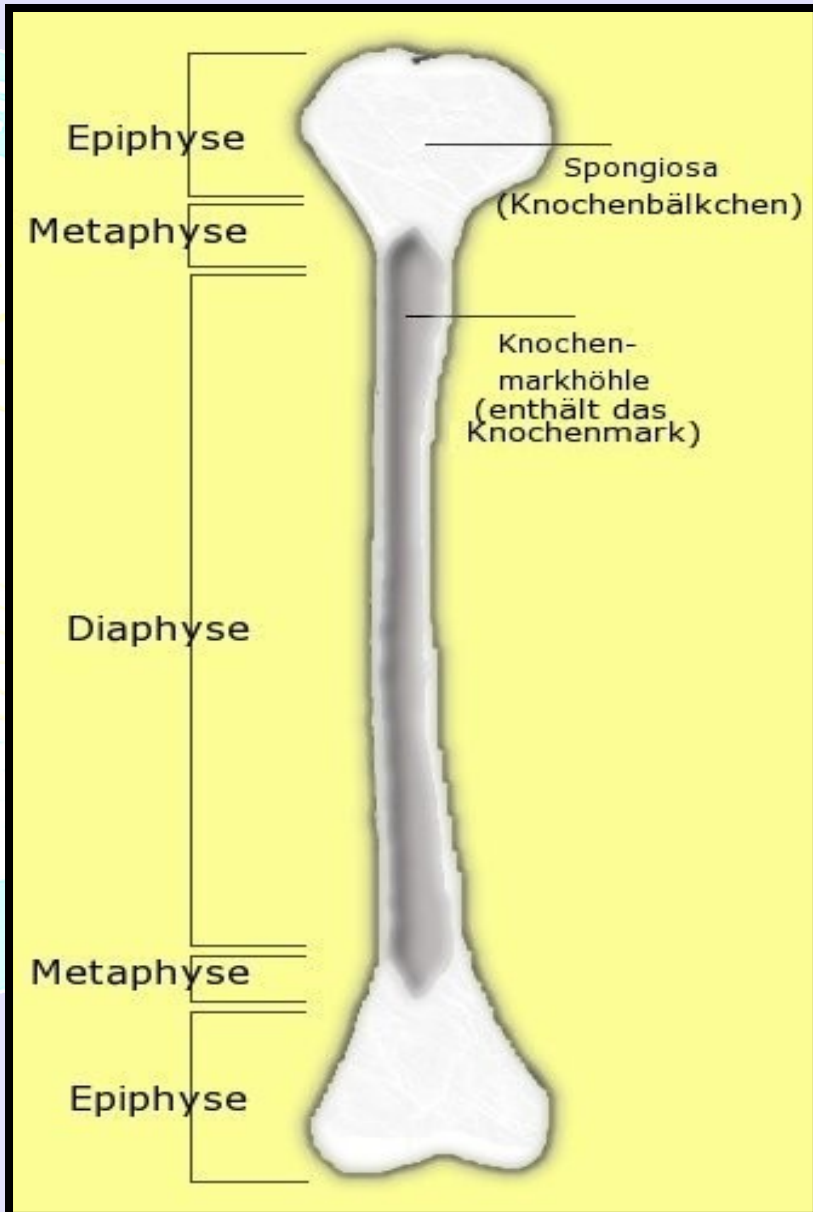
PERIOSTEUM

- The exterior of bones (except where they interact with other bones through ***joints***) is covered by the ***periosteum.***
- **Consists of*:***
  - External **fibrous** layer
  - Internal ***osteogenic*** layer.
- The periosteum is richly supplied with blood, lymph and nerve vessels, attaching to the bone itself through **Sharpey's fibres.**



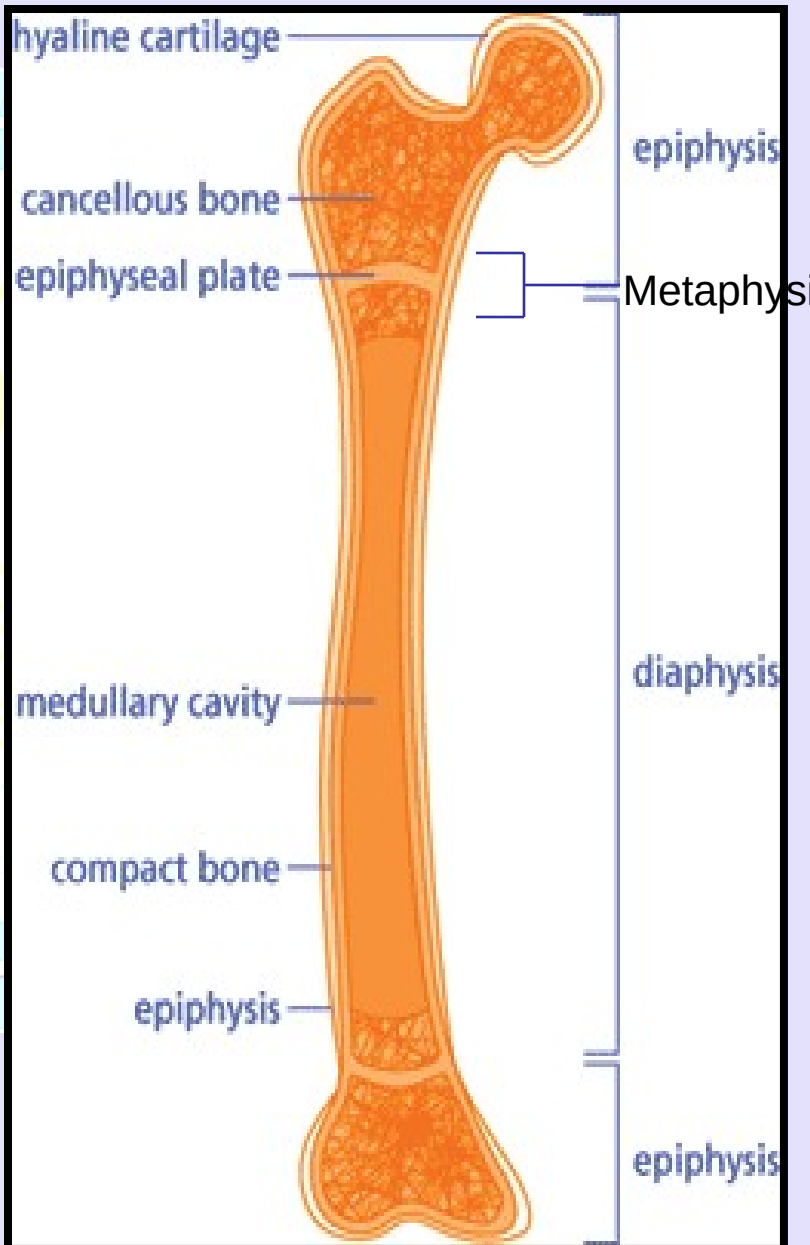
PARTS OF YOUNG BONE

- **Diaphysis** It is shaft or long main portion.
- **Epiphysis** It is end. The two ends together are called the epiphyses. Each epiphysis is covered with articular cartilage.
- **Metaphysis** It is the region of mature bones where the diaphysis meets the epiphysis.
- **Epiphyseal plate** In a growing bone, the epiphyseal plate is formed of hyaline cartilage. Under the influence of growth hormone, the plate continues to grow, giving length to the bone. At beginning of puberty, the epiphyseal plate is slowly lost.



YOUNG LONG BONE

In the young bone the Epiphysis are not united with Diaphysis but the bone is in the state of development and not ossified. The epiphyses and diaphysis are separated by **Epiphyseal** cartilage

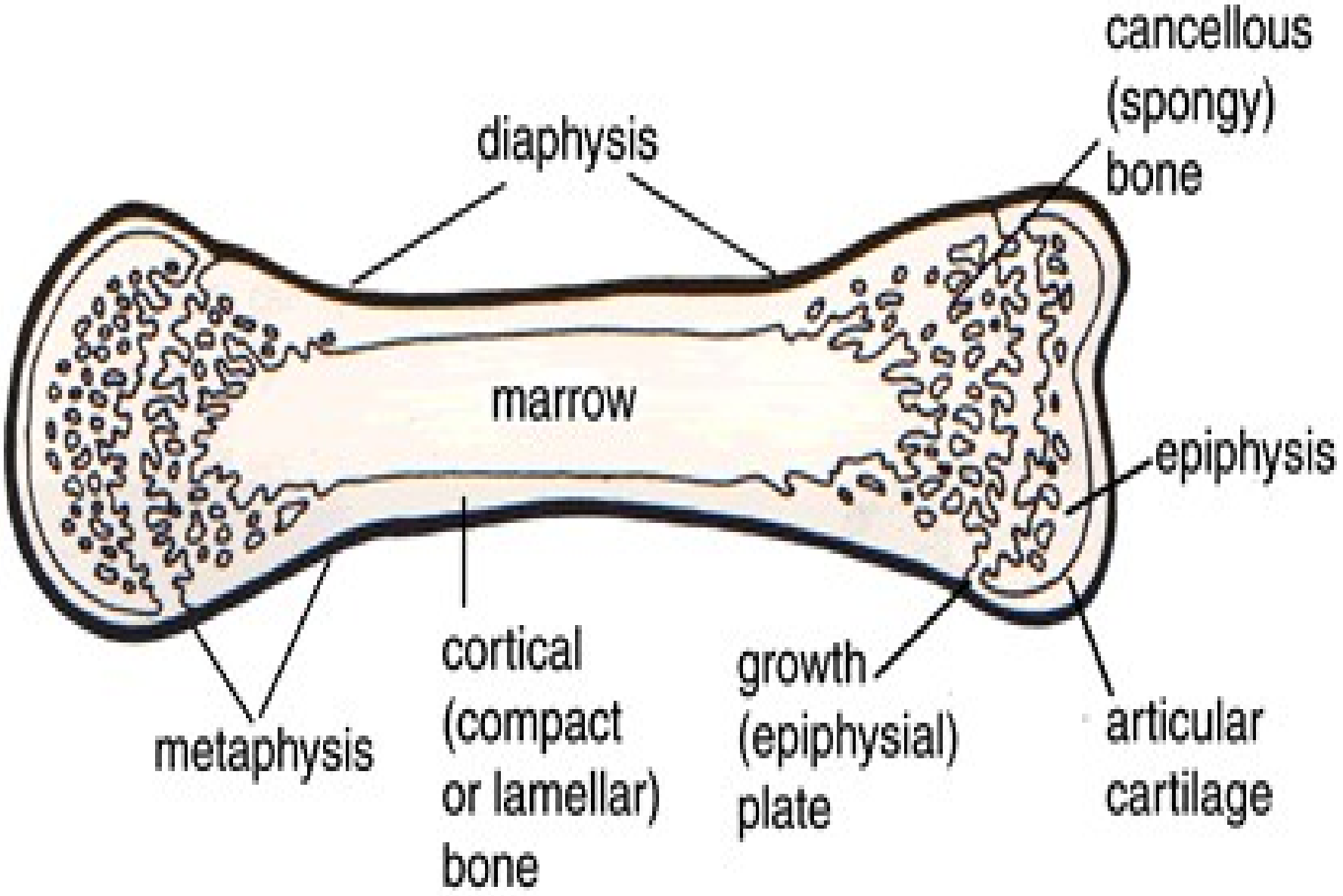


DIFFERENCE BETWEEN ADULT AND YOUNG BONE





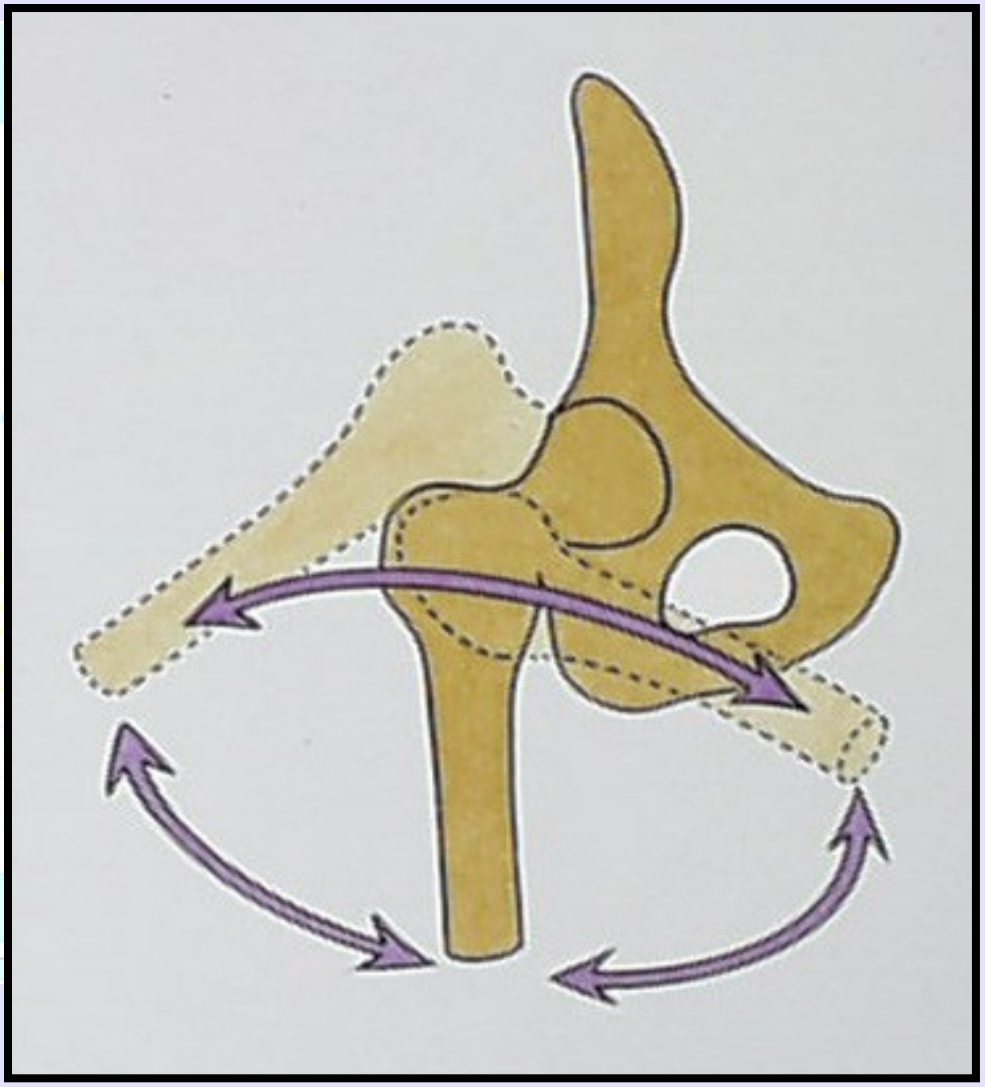
Long bone during growth



FUNCTIONS OF THE SKELETON

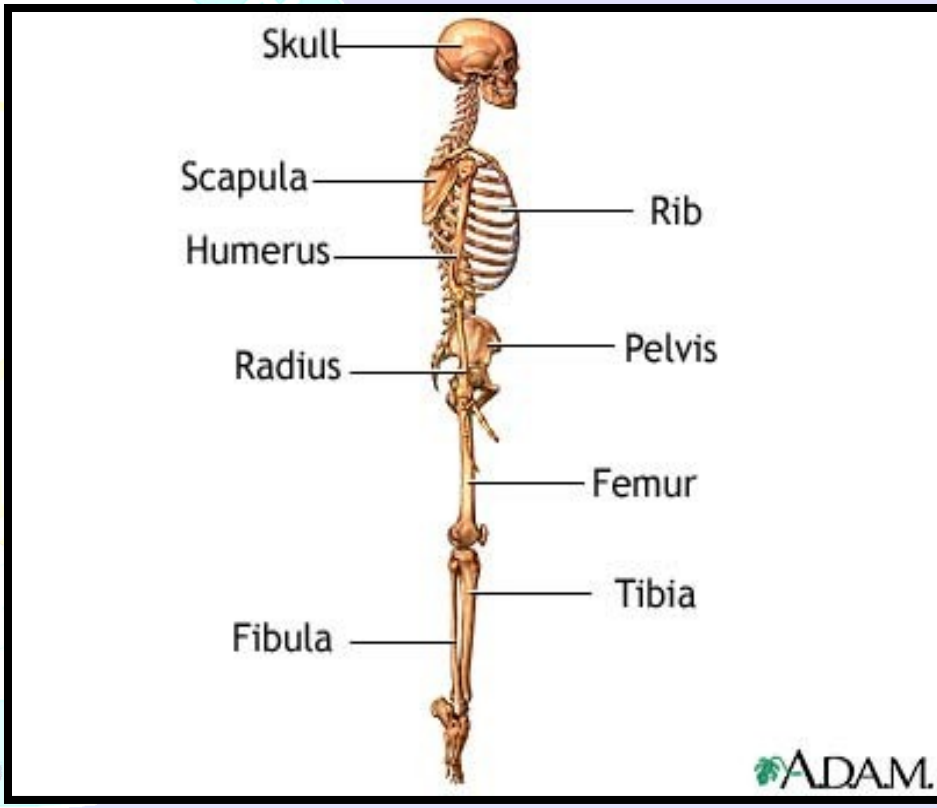
Movement

- If the skeleton did not have joints, it would be impossible to move. Bones are linked at the joints and muscles allow them to move. The way the bones grow affects the movement in the joints.



FUNCTIONS OF THE SKELETON

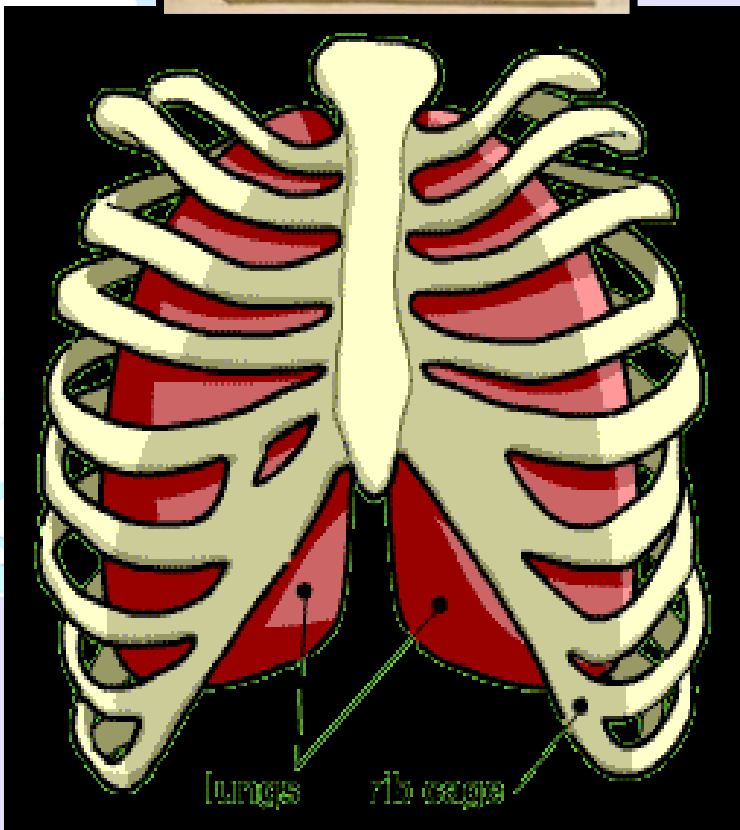
SUPPORT: Without a skeleton we would be like jelly and flop all over the floor. In fact, one could argue that our skeleton is an important evolutionary trait that has helped our species progress. Our skeleton provides the body with shape and forms a frame under the skin. The body is held in position by muscles which are firmly attached to the bones.



FUNCTION OF THE SKELETON

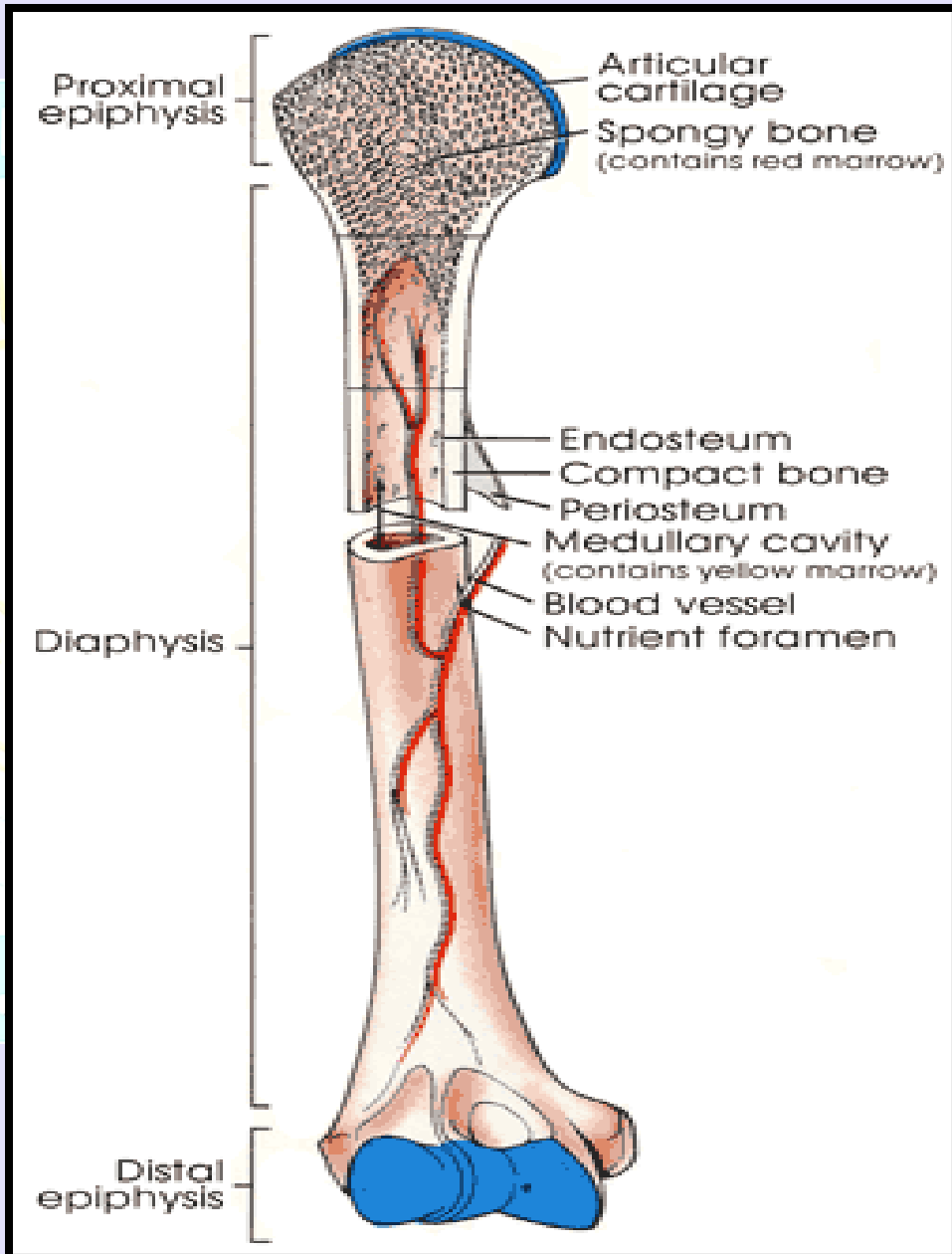


- ***Protection.***
  - Vital organs are protected from damage by the different bones of the skeleton. The brain is protected by the cranium, the heart and lungs are surrounded by the ribs and the sternum, and the spinal cord passes through the centre of the Vertebral Column
- **Muscles Attachment**



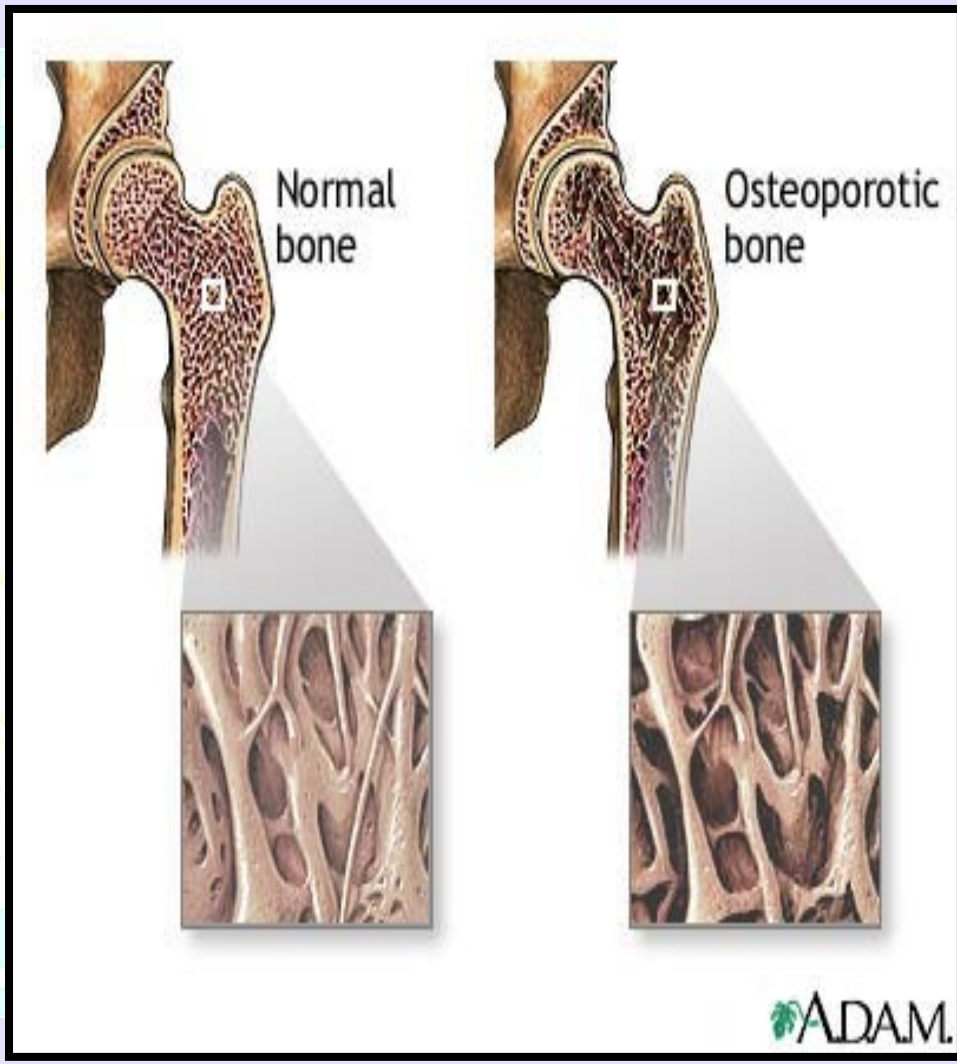
BLOOD PRODUCTION

- Blood cells are made in the red bone marrow in the centre of certain bones of the skeleton. The main sites of blood cell production are the pelvic girdle, ribs, sternum and the vertebrae.
- Bones are not completely solid, as they would be too heavy to carry around. Blood vessels feed the centre of the bones, and calcium is stored and released to the tissues which require it.



STORE FOR MINERALS

The main mineral that is stored in the skeleton is calcium. The body needs calcium to harden the bones. This hardening process is called **ossification** and happens throughout childhood. The skeleton is also continually replacing itself throughout our adult life. Calcium is transported in the bloodstream to the sites of bone growth or where replacement is needed (at an injured bone).



REFERENCES

- **Clinically Oriented Anatomy**

  Keith L. Moore

  6th edition

  Chapter 01

  Pages no: 19-22



THANK YOU